

CONSULTAȚII LMC
2 EXERCITII
RECAPITULATIVE

Exerc. 1: Fie $B = (B, \vee, \wedge, \subseteq, \bar{}, 0, 1)$
o algebra Boole,

① Fie $R = \{(x, \bar{x}) \mid x \in B\} \subseteq B^2$.
Dem. că:

- (a) R e reflexivă $\Leftrightarrow B$ e neviduie
- (b) R e simetric,
- (c) R e tranzitivă $\Leftrightarrow B$ e trivial,

② Fie $x, y \in B$. Dem. că:

- (d) $[x] \cap [y] = [x \vee y]$;
- (e) $[[x] \cup [y]] = [x \wedge y]$.

③ Fie $\mathcal{I} \neq \emptyset$ o mulțime
 $(\forall i \in \mathcal{I})(\emptyset \neq X_i \subseteq B)$. Dem. că:

$$[\bigcup_{i \in \mathcal{I}} [X_i]] = [\bigcup_{i \in \mathcal{I}} X_i],$$

Pr. ② e ③, seguinte ex:

$$\begin{aligned}
 & (\forall \alpha \in B) (\ulcorner \alpha \urcorner = \{a \in B \mid \alpha \leq a\}) \\
 & (\forall \emptyset \neq A \subseteq B) (\ulcorner A \urcorner = \{a \in B \mid (\exists n \in \mathbb{N}^*) (\exists \alpha_1, \dots, \alpha_n \in A) (\alpha_1 \wedge \dots \wedge \alpha_n \leq a)\}) \\
 & \Rightarrow (\forall n \in \mathbb{N}^*) (\forall \alpha_1, \dots, \alpha_n \in B) \\
 & (\ulcorner \{\alpha_1, \dots, \alpha_n\} \urcorner = \ulcorner \alpha_1 \wedge \dots \wedge \alpha_n \urcorner)
 \end{aligned}$$

Pr. ③, seguinte ex:

$$\begin{aligned}
 & (\forall X \subseteq Y \subseteq B) (\ulcorner X \urcorner = \ulcorner Y \urcorner) \\
 & \text{ex: } \left(\begin{array}{l} \ulcorner Y \urcorner \supseteq Y \supseteq X \\ \ulcorner X \urcorner \in \mathcal{F}(B) \end{array} \right) \xrightarrow{\text{pr. ②}} \ulcorner Y \urcorner = \ulcorner X \urcorner
 \end{aligned}$$

$$\begin{aligned}
 & \textcircled{1} \quad (a) \xrightarrow[\mathcal{F}(B)]{u} R \text{ e reflexiva} \Rightarrow \\
 & \Rightarrow 0R0 \Leftrightarrow 0 \neq \overline{0} \Leftrightarrow 0 \neq 1 \Leftrightarrow B \\
 & \text{e neutro.}
 \end{aligned}$$

$$\begin{aligned}
 & \xleftarrow[\mathcal{F}(B)]{u} B \text{ e neutro.} \Leftrightarrow 0 \neq 1 \\
 & \text{Pr. obs co } R \text{ m e reflexiva} \\
 & \Leftrightarrow (\exists x \in B) (xRx) \Leftrightarrow (\exists x \in B) (x = \overline{x}) \Rightarrow
 \end{aligned}$$

$$\Rightarrow \left\{ \begin{array}{l} 1 = x \vee \bar{x} = x \vee x = x \\ 0 = x \wedge \bar{x} = x \wedge x = x \end{array} \right\} \Rightarrow 0 = 1 \rightarrow \text{contradiction}$$

$\Rightarrow R$ e reflexiva.

(b) $(\forall x, y \in B) (x R y \Leftrightarrow y = \bar{x} \Leftrightarrow \overline{y} = \overline{\bar{x}} \Leftrightarrow \overline{y} = x \Leftrightarrow y R x) \Rightarrow R$ e simétrica.

(c) $\underbrace{B}_{\neq \emptyset} \subseteq B$ e transitiva, $\Leftrightarrow 0 = 1 \Leftrightarrow$

$$\Leftrightarrow B = 0 \xrightarrow{\text{by } R} R = \{(0,0)\} = \{(0,1)\} = \{(1,0)\} = B^2 \rightarrow \text{transitiva.}$$

$\underbrace{B}_{\neq \emptyset} \subseteq R \rightarrow \text{transitiva.}$

$$\begin{array}{l} 0 R 0 \Leftrightarrow 0 R 1, \\ 1 R 1 \Leftrightarrow 1 R 0. \end{array}$$

$$\begin{array}{l} \Rightarrow 0 R 0 \Leftrightarrow \\ \Leftrightarrow 0 = \overline{0} \Leftrightarrow \\ \Leftrightarrow 0 = 1 \Leftrightarrow B \\ \text{e transitiva.} \end{array}$$

(2) (d) Se $a \in B$, either, first

$$a \in [x] \cap [y] \Leftrightarrow \begin{cases} a \in [x] \\ a \in [y] \end{cases} \Leftrightarrow \begin{cases} x \leq a \\ y \leq a \end{cases} \Leftrightarrow x \vee y \leq a$$

$$\Leftrightarrow x \vee y \leq a \Leftrightarrow a \in [x \vee y] \Rightarrow$$

$$\Rightarrow [x] \cap [y] = [x \vee y]$$

$$(e) [x \wedge y] = [x, y],$$

$$F := [[x] \cup [y]], \quad F = [[x, y]]$$

$$\begin{aligned} x \in [x] &\subseteq [x] \cup [y] \subseteq [x] \cap [y] = [x] \\ y \in [y] &\subseteq [x] \cup [y] \subseteq [x] \cap [y] = [x] \end{aligned} \Rightarrow F = [[x] \cup [y]] = F$$

$$\Rightarrow F \subseteq [x, y]. \quad (*)$$

$$F \in F(B). \quad (**)$$

$$\# G \in F(B), \text{ s.t. }, G \subseteq [x, y] \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} G \subseteq x \Rightarrow G \subseteq [x] \\ G \subseteq y \Rightarrow G \subseteq [y] \end{cases} \Leftrightarrow G \subseteq [x] \cap [y] \Rightarrow$$

$$\Rightarrow G \subseteq [[x] \cup [y]] = F. \quad (***)$$

$$(*) \quad (***) \quad (**) \Rightarrow F = [[x, y]]$$

$$\Rightarrow [[x] \cup [y]] = [x \vee y]$$

$$\textcircled{3} (\forall i \in I) (X_i \subseteq [X_i]) \Rightarrow \bigcup_{i \in I} X_i \subseteq \bigcup_{i \in I} [X_i] \\ \Rightarrow [\bigcup_{i \in I} X_i] \subseteq [\bigcup_{i \in I} [X_i]]$$

$$\text{For } a \in [\bigcup_{i \in I} [X_i]], \Leftrightarrow (\exists n \in \mathbb{N}^*) \\ (\exists \alpha_1, \dots, \alpha_n \in \bigcup_{i \in I} [X_i]) (\alpha_1 \wedge \dots \wedge \alpha_n \leq a) \quad \textcircled{I}$$

$$(\forall k \in \overline{1, n}) (\alpha_k \in \bigcup_{i \in I} [X_i]) \Leftrightarrow \\ \Leftrightarrow (\forall k \in \overline{1, n}) (\exists i_k \in I) (\alpha_k \in [X_{i_k}])$$

$$\Leftrightarrow (\exists m_k \in \mathbb{N}^*) (\exists \beta_{k,1}, \beta_{k,2}, \dots, \beta_{k,m_k} \in X_{i_k}) \left(\bigwedge_{j=1}^{m_k} \beta_{k,j} \leq \alpha_k \right) \quad \textcircled{II}$$

$\subseteq \bigcup_{i \in I} X_i$

$$\textcircled{I}, \textcircled{II} \Rightarrow \bigwedge_{k=1}^n \bigwedge_{j=1}^{m_k} \beta_{k,j} \leq \bigwedge_{k=1}^n \alpha_k \leq a$$

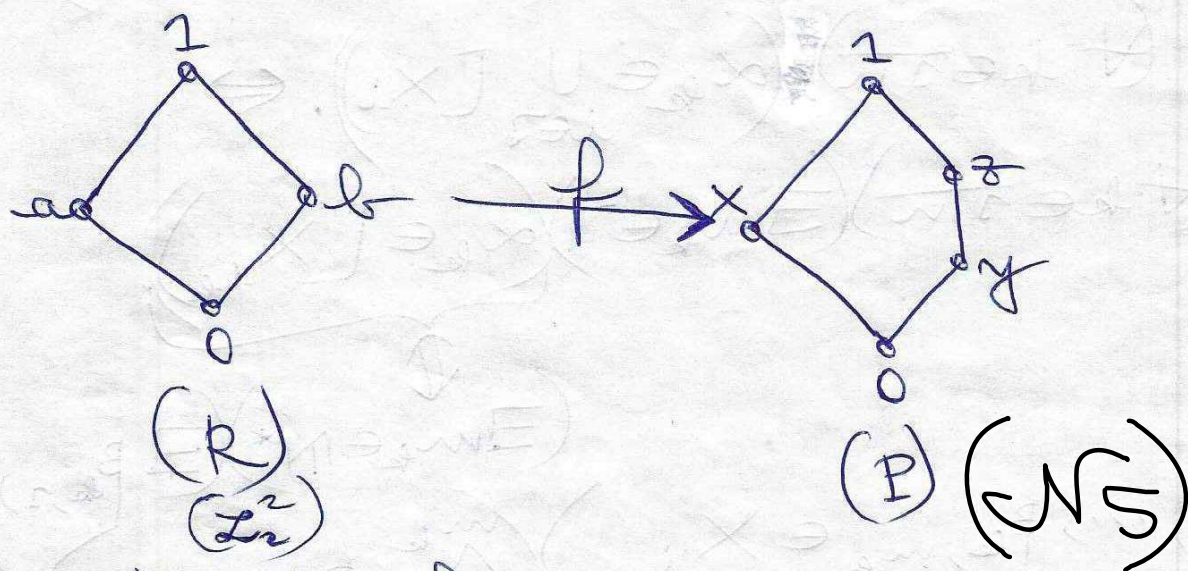
$\bigcup_{i \in I} X_i$

$$\Rightarrow a \in [\bigcup_{i \in I} X_i] \Rightarrow [\bigcup_{i \in I} [X_i]] \subseteq [\bigcup_{i \in I} X_i]$$

$$\Rightarrow [\bigcup_{i \in I} [X_i]] = [\bigcup_{i \in I} X_i]$$

Exerci: Fie $R := \text{rombul}$, $P := \text{pentagonul}$ și $f: R \rightarrow P$ un morfism de lattice marginite. Demonstrați că:
 $f(R) \neq \{0, 1\} \Leftrightarrow f$ e injectiv.

REZOLVARE:



Fie $R = \{0, a, b, 1\}$ și $P = \{0, x, y, z, 1\}$, cu elementele notate ca în figura de mai sus.

$\Leftarrow$: Ipoteza acestei implicații este că f e injectiv.

Pp. prin absurd că $f(R) = \{0, 1\} \Rightarrow |R| = 4 > 2 = |\{0, 1\}| = |f(R)| \Rightarrow f$ nu poate fi injectiv. $\Rightarrow f(R) \neq \{0, 1\}$.

-4-

 $\Rightarrow$

Ipoteza acestei implicații este
 că $f(\mathbb{R}) \neq \{0, 1\}$.

f e morfism de lotici marginite. $\Rightarrow$
 $\Rightarrow f(0) = 0$ și $f(1) = 1. \Rightarrow \{0, 1\} = f(\mathbb{R})$

$\Rightarrow \{0, 1\} \subsetneq f(\mathbb{R}). \Rightarrow \exists \beta \in f(\mathbb{R}) \setminus \{0, 1\}$.

$\beta \in f(\mathbb{R}) \Leftrightarrow (\exists \alpha \in \mathbb{R}) (f(\alpha) = \beta)$

$f(0) = 0$ și $\beta \neq 0.$

$f(1) = 1$ și $\beta \neq 1.$

$\Rightarrow \alpha \in \mathbb{R} \setminus \{0, 1\} = \{a, b\}.$

Pp. că $a = \alpha$, cazul
 se tratează analog, $b = \alpha$

Așadar, $f(a) = \beta \in \mathbb{R} \setminus \{0, 1\} =$
 $= \{x, y, z\}.$

Pp. prin absurd că f nu e
 injectiv.

$f(0) = 0, f(1) = 1$ și $f(a) = \beta$
 sunt 3 cazuri distincte.

$$\Rightarrow f(b) \in \{f(a), f(1), f(0)\} = \{0, 1, \beta\}.$$

Case 1: $f(b) = 0.$

$$1 = f(a \vee b) = f(a) \vee f(b) \stackrel{f(b)=0}{=} \beta \vee 0 = \beta. \Rightarrow \beta = 1 \rightarrow \text{X}_0.$$

Case 2: $f(b) = 1.$

$$0 = f(a \wedge b) = f(a) \wedge f(b) \stackrel{f(b)=1}{=} \beta \wedge 1 = \beta. \Rightarrow \beta = 0. \rightarrow \text{X}_0.$$

Case 3: $f(b) = \beta.$

$$0 = f(a \wedge b) = f(a) \wedge f(b) \stackrel{f(b)=\beta}{=} \beta \wedge \beta = \beta. \Rightarrow \beta = 0 \rightarrow \text{X}_0.$$

$\Rightarrow f$ is injective.

Exerc.: For $\alpha, \varphi, \psi, \chi \in E$, a.s.

$$\alpha = (\varphi \rightarrow \neg(\neg\psi \rightarrow \chi)) \leftrightarrow ((\psi \rightarrow \neg\varphi) \wedge (\chi \rightarrow \neg\varphi)).$$

Demonstrate it:

$$\vdash \alpha \text{ " } \begin{cases} (a) & \text{algebraic, SXIII, '12-'13} \\ (b) & \text{semantic.} \end{cases}$$

(a) Fie $x := \varphi$, $y := \psi$ și $z := \neg x$
 în $E/\sim$.

$$\begin{aligned}
 \Rightarrow \alpha &= (\varphi \rightarrow (\overline{\psi} \rightarrow \neg x)) \leftrightarrow \\
 &\leftrightarrow ((\overline{\psi} \rightarrow \overline{\varphi}) \wedge (\neg x \rightarrow \overline{\varphi})) = \\
 &= (x \rightarrow (\overline{y} \rightarrow z)) \leftrightarrow ((y \rightarrow x) \vee (z \rightarrow x)) \leftrightarrow \\
 &= (x \rightarrow (\overline{y} \vee z)) \leftrightarrow ((\overline{y} \vee x) \vee (z \vee x)) = \\
 &= (\overline{x} \vee (\overline{y} \vee z)) \leftrightarrow ((\overline{y} \vee z) \vee \overline{x}) \quad (\text{de asociativitate}) \\
 &= (\overline{x} \vee (\overline{y} \vee z)) \leftrightarrow (\overline{x} \vee (\overline{y} \wedge z)) = 1
 \end{aligned}$$

fiind o echivalență booleană
 între doi termeni egali,

$$\text{Deci } \alpha = 1. \Leftrightarrow \vdash \alpha.$$

(b) Demonstrăm că $\models \alpha$.

Fie $h: V \rightarrow L_2$, arbitrară, fixată.

$$\begin{aligned}
 \tilde{h}(\alpha) &= (\tilde{h}(\varphi) \rightarrow (\overline{\tilde{h}(\psi)} \rightarrow \tilde{h}(\neg x))) \leftrightarrow \\
 &\leftrightarrow ((\tilde{h}(\psi) \rightarrow \tilde{h}(\varphi)) \wedge (\tilde{h}(\neg x) \rightarrow \tilde{h}(\varphi))),
 \end{aligned}$$

De aici putem continua cu un

calcul boolean, ca la punctul (a);
sau putem proceda astfel: notam

$$\beta := \varphi \rightarrow \neg(\neg\psi \rightarrow \chi) \in E$$

$$\gamma := (\psi \rightarrow \neg\varphi) \wedge (\chi \rightarrow \neg\varphi) \in E. \quad \text{și}$$

$$\tilde{h}(\alpha) = \tilde{h}(\beta) \leftrightarrow \tilde{h}(\gamma),$$

$$\text{exemplu: } \tilde{h}(\alpha) = 1 \Leftrightarrow \tilde{h}(\beta) = \tilde{h}(\gamma),$$

Caz 1: $\tilde{h}(\beta) = 0 \Leftrightarrow$

$$\Leftrightarrow \tilde{h}(\varphi) \rightarrow \overline{(\tilde{h}(\psi) \rightarrow \tilde{h}(\chi))} = 0 \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} \tilde{h}(\varphi) = 1 \\ \text{și} \end{cases}$$

$$\overline{\tilde{h}(\psi) \rightarrow \tilde{h}(\chi)} = 0 \Leftrightarrow$$

$$\Leftrightarrow \overline{\tilde{h}(\psi) \rightarrow \tilde{h}(\chi)} = 1 \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} \tilde{h}(\psi) = 0 \Leftrightarrow \tilde{h}(\psi) = 1, \\ \text{sau} \\ \tilde{h}(\chi) = 1. \end{cases}$$

$$\text{Atadar: } \tilde{h}(\beta) = 0 \Leftrightarrow \begin{cases} \tilde{h}(\varphi) = \tilde{h}(\psi) = 1 \\ \text{sau} \\ \tilde{h}(\varphi) = \tilde{h}(\chi) = 1 \end{cases} \Leftrightarrow$$

— 6 —

$$\Leftrightarrow \begin{cases} \underbrace{\tilde{L}(\psi)}_{\text{sur}} \rightarrow \overline{\tilde{L}(\varphi)} = 0 \\ \tilde{L}(x) \rightarrow \overline{\tilde{L}(\varphi)} = 0 \end{cases} \Leftrightarrow$$

Ex 2: $\tilde{L}(p) = 1 \xrightarrow{\text{coz}} \tilde{L}(y) = 1$

$$\Rightarrow \models \alpha, \stackrel{\text{TC}}{\iff} \vdash \alpha.$$

RESOLVARE:

$\Gamma \subseteq E$ $\exists \varphi, \psi, \chi \in E, \text{ s.t.}$
 $\int \Gamma \cup \{\varphi\} \vdash \chi$ (s.t.) $\int \Gamma \cup \{\varphi\} \models \chi$ (1)
 $\int \Gamma \cup \{\psi\} \vdash \chi$ $\iff$ $\int \Gamma \cup \{\psi\} \models \chi$ (2)

$$\Gamma \cup \{\varphi \vee \psi\} \models \gamma. \quad \#$$

Fie $h: V \rightarrow L_2$, e.i.d.

$$h \models \Gamma \cup \{\varphi \vee \psi\} \Leftrightarrow \begin{cases} h \models \Gamma & (*) \\ \tilde{h}(\varphi \vee \psi) = 1 \end{cases}$$

$$\Leftrightarrow \tilde{h}(\varphi) \vee \tilde{h}(\psi) = 1.$$

$$\tilde{h}(\varphi), \tilde{h}(\psi) \in L_2 = \{0, 1\}.$$

$$\Rightarrow \tilde{h}(\varphi) = 1 \text{ sau } \tilde{h}(\psi) = 1.$$

Caz 1: Dacă $\tilde{h}(\varphi) = 1 \xRightarrow{(*)}$

$$\xRightarrow{(*)} h \models \Gamma \cup \{\varphi\} \xRightarrow{(1)} \tilde{h}(\varphi) = 1.$$

Caz 2: Dacă $\tilde{h}(\psi) = 1 \xRightarrow{(*)}$

$$\xRightarrow{(*)} h \models \Gamma \cup \{\psi\} \xRightarrow{(2)} \tilde{h}(\psi) = 1.$$

$$\Rightarrow \Gamma \cup \{\varphi \vee \psi\} \models \gamma. \quad \Leftrightarrow$$

$$\Leftrightarrow \Gamma \cup \{\varphi \vee \psi\} \vdash \gamma.$$

ALTĂ METODĂ LA (b) LA
EXERCITIUL ENUNȚAT PE VERSO-UL
PĂGINII 4:

Amintesc că, în orice algebra
Boole B , pt. orice $a \in B$, avem:

$$\begin{aligned}
 a &= a, \Leftrightarrow a \leftrightarrow a = 1, & (*) \\
 0 \leq a &\Leftrightarrow 0 \rightarrow a = 1, & (*) \\
 a \leq 1 &\Leftrightarrow a \rightarrow 1 = 1, & (*) \\
 a \rightarrow 0 &= \overline{a} \vee 0 = \overline{a}, & (***) \\
 1 \rightarrow a &= \overline{1} \vee a = 0 \vee a = a, & (***)
 \end{aligned}$$

Se $h: V \rightarrow L_2 = \{0, 1\}$ arbitrária fixada.

$$\begin{aligned}
 h(\alpha) &= (h(\varphi) \rightarrow (h(\psi) \rightarrow h(\chi))) \\
 &\Leftrightarrow ((h(\psi) \rightarrow h(\varphi)) \wedge (h(\chi) \rightarrow h(\varphi)))
 \end{aligned}$$

Caso 1: Se $h(\varphi) = 0 \Rightarrow \overline{h(\varphi)} = \overline{0} = 1$

$$\begin{aligned}
 &= 1 \Rightarrow h(\alpha) = (0 \rightarrow (h(\psi) \rightarrow h(\chi))) \\
 &\Leftrightarrow ((h(\psi) \rightarrow 1) \wedge (h(\chi) \rightarrow 1)) \quad (**), (***) \\
 &= 1 \Leftrightarrow (1 \wedge 1) = 1 \Leftrightarrow 1 \stackrel{(*)}{=} 1.
 \end{aligned}$$

Caso 2: Se $h(\varphi) = 1 \Rightarrow \overline{h(\varphi)} = \overline{1} = 0$

$$\begin{aligned}
 &= 0 \Rightarrow h(\alpha) = (1 \rightarrow (h(\psi) \rightarrow h(\chi))) \\
 &\Leftrightarrow ((h(\psi) \rightarrow 0) \wedge (h(\chi) \rightarrow 0)) \quad (**), (***) \\
 &= \overline{(h(\psi) \rightarrow h(\chi))} \Leftrightarrow \overline{(h(\psi) \wedge h(\chi))} = \\
 &\quad = \overline{h(\psi)} \vee \overline{h(\chi)} = \overline{h(\psi)} \vee \overline{h(\chi)} = \\
 &\quad = \overline{h(\psi)} \wedge \overline{h(\chi)} \\
 &= (\overline{h(\psi)} \wedge \overline{h(\chi)}) \Leftrightarrow (\overline{h(\psi)} \wedge \overline{h(\chi)}) \stackrel{(*)}{=} 0 \\
 &\Rightarrow (\forall h: V \rightarrow L_2) (h \models \alpha) \Leftrightarrow \\
 &\stackrel{def}{=} \models \alpha \stackrel{def}{=} \vdash \alpha.
 \end{aligned}$$

Exercițiu: Să se demonstreze semantic
corectitudinea regulii de deducție: pentru
orice $\Sigma_1, \Sigma_2, \Sigma_3 \in \mathcal{P}(\Theta)$ și orice $\varphi, \psi, \chi \in E$:

$$\frac{\Sigma_1 \cup \{\varphi\} \vdash \psi, \Sigma_2 \cup \{\psi \wedge \chi\} \vdash \varphi, \Sigma_3 \cup \{\chi\} \vdash \chi}{\Sigma_1 \cup \Sigma_2 \cup \Sigma_3 \vdash \varphi \leftrightarrow (\psi \wedge \chi)}.$$

REZOLVARE: Fie $\Sigma_1, \Sigma_2, \Sigma_3 \in \mathcal{P}(\Theta)$
și $\varphi, \psi, \chi \in E$ a. d.

cu loc deductiv, sintactice:

$$\left\{ \begin{array}{l} \Sigma_1 \cup \{\varphi\} \vdash \psi \iff \Sigma_1 \cup \{\varphi\} \models \psi; (1) \\ \Sigma_2 \cup \{\psi \wedge \chi\} \vdash \varphi \iff \Sigma_2 \cup \{\psi \wedge \chi\} \models \varphi; (2) \\ \Sigma_3 \cup \{\chi\} \vdash \chi \iff \Sigma_3 \cup \{\chi\} \models \chi. (3) \end{array} \right.$$

$$\text{Fie } h: V \rightarrow \{0, 1\} \text{ a. d. } h \models \Sigma_1 \cup \Sigma_2 \cup \Sigma_3 \\ \iff h \models \Sigma_1, h \models \Sigma_2, h \models \Sigma_3.$$

Caz 1: Dacă $\mathcal{V}(\varphi) = 1$ ($h \models \Sigma_1$)

$$\Rightarrow h \models \Sigma_1 \cup \{\varphi\} \xrightarrow{(1)} \mathcal{V}(\psi) = 1 \xrightarrow{(2)} \mathcal{V}(\psi \wedge \chi) = 1$$

$$\Rightarrow h \models \Sigma_3 \cup \{\chi\} \xrightarrow{(3)} \mathcal{V}(\chi) = 1 \Rightarrow$$

$$\Rightarrow \mathcal{V}(\varphi \leftrightarrow (\psi \wedge \chi)) = \mathcal{V}(\varphi) \leftrightarrow (\mathcal{V}(\psi \wedge \chi)) = \\ = 1 \leftrightarrow (1 \wedge 1) = 1 \leftrightarrow 1 = 1.$$

Caz 2: Dacă $\mathcal{V}(\varphi) = 0$:

$$\text{Presupunem prin absurd că } \mathcal{V}(\psi \wedge \chi) = 1 \Rightarrow \\ \xrightarrow{(2)} h \models \Sigma_2 \cup \{\psi \wedge \chi\} \xrightarrow{(2)} \mathcal{V}(\varphi) = 1 \nRightarrow$$

$$\Rightarrow \mathcal{V}(\psi \wedge \chi) = 0 \Rightarrow \mathcal{V}(\varphi \leftrightarrow (\psi \wedge \chi)) = \\ = \mathcal{V}(\varphi) \leftrightarrow \mathcal{V}(\psi \wedge \chi) = 0 \leftrightarrow 0 = 1.$$

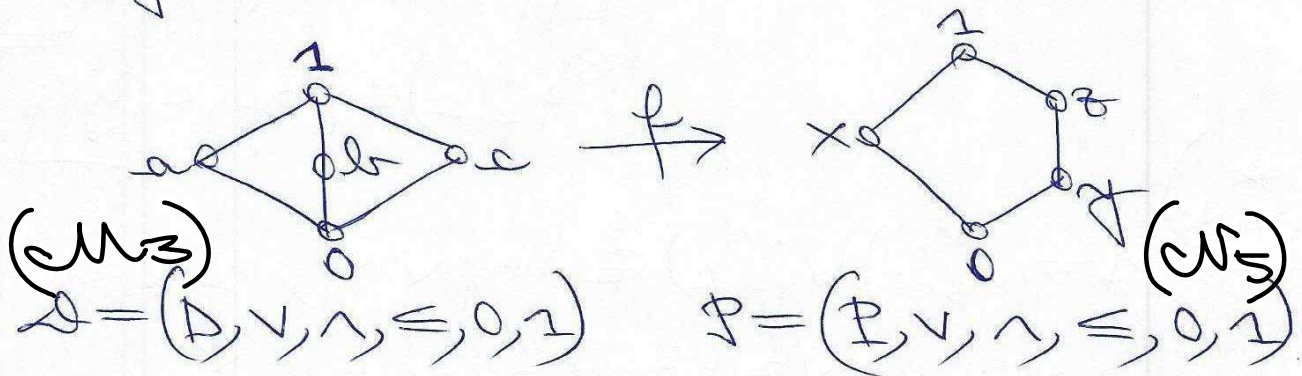
$$\Rightarrow \Sigma_1 \cup \Sigma_2 \cup \Sigma_3 \models \varphi \leftrightarrow (\psi \wedge \chi) \xrightarrow{(1)} \Sigma_1 \cup \Sigma_2 \cup \Sigma_3 \models \varphi \leftrightarrow (\psi \wedge \chi).$$

- 8 -

Exercitiu: să se demonstreze că
nu există morfisme de latici
mărginite de la diamant la pentagon.

CONSULTATII
LMC
că

REZOLVARE: Notăm elementele diamantului
și pentagonului ca în următoarele
diagrame Hasse:



Presupunem prin absurd că
 $f: D \rightarrow P$ este un morfism de
latici mărginite. Rezultă:

$$f(0) = 0 \text{ și } f(1) = 1;$$

$$\begin{cases} a \vee b = 1 \Rightarrow f(a) \vee f(b) = f(a \vee b) = f(1) = 1 \\ a \wedge b = 0 \Rightarrow f(a) \wedge f(b) = f(a \wedge b) = f(0) = 0 \end{cases} \quad (*)$$

Analog rezultă: $\begin{cases} f(a) \vee f(c) = 1; \\ f(a) \wedge f(c) = 0; \end{cases} \quad (**)$

și: $\begin{cases} f(b) \vee f(c) = 1; \\ f(b) \wedge f(c) = 0. \end{cases} \quad (***)$

Caz 1: $f(a) = 0 \xrightarrow{(*)} f(b) = 1 \xrightarrow{(***)} f(c) = 1$

$$\Rightarrow 0 = f(b) \wedge f(c) = 1 \wedge 1 = 1. \quad \times$$

Caz 2: $f(a) = 1, \begin{matrix} (*) \\ \swarrow \end{matrix} \left. \begin{matrix} f(b) = 0 \\ f(c) = 0 \end{matrix} \right\} \begin{matrix} (*) \\ \swarrow \end{matrix}$

$$\Rightarrow 1 = f(b) \vee f(c) = 0 \vee 0 = 0. \quad \times$$

Caz 3: $f(a) = x, \begin{matrix} (*) \\ \swarrow \end{matrix} \left. \begin{matrix} f(b) \in \{x, z\} \\ f(c) \in \{x, z\} \end{matrix} \right\} \begin{matrix} (*) \\ \swarrow \end{matrix}$

$$\begin{matrix} (*) \\ \swarrow \end{matrix} \Rightarrow 1 = f(b) \vee f(c) \in \{x \vee x, z \vee z\} = \{x, z\} \neq 1. \quad \times$$

Caz 4: $f(a) \in \{x, z\}, \begin{matrix} (*) \\ \swarrow \end{matrix} \left. \begin{matrix} f(b) = x \\ f(c) = x \end{matrix} \right\} \begin{matrix} (*) \\ \swarrow \end{matrix}$

$$\begin{matrix} (*) \\ \swarrow \end{matrix} \Rightarrow 1 = f(b) \vee f(c) = x \vee x = x. \quad \times$$

$\Rightarrow$ Nu exista morfisme de latex marginite de la $\mathcal{D}$ la $\mathcal{P}$.

Exercitiu temă:

Demonstrati ca morfismele de latex marginite

(notatiile consacrate)

$Hh: \mathcal{P} \rightarrow \mathcal{D}$ sunt:

$(\mathcal{C}^{\mathcal{N}}_5) (\mathcal{C}^{\mathcal{M}}_3)$

u	0	x	y	z	1
$h(u)$	0	0	1	1	1
	0	1	0	0	1
	0	0	0	1	1
	0	0	1	0	1
	0	1	1	0	1
	0	1	0	1	1
	0	0	1	1	1
	0	0	0	1	1

ca sa se noteze din exercitiul anterior.

Pe elementele diagramelor Hase